



Build a Floating House

My name _____

Class _____

Date _____



How can we build a house that floats safely in a flood?

Bangkok has lots of water, and floods are coming more often. Some families live in houses that float! Let's build one and test it.



My kit tick when you have it

- ☐ 4 empty plastic bottles with lids
- ☐ Stiff card or a plastic tray for the floor
- ☐ Strong tape
- ☐ LEGO bricks for the house
- ☐ A big tub of water
- ☐ Towels for spills
- ☐ 2 more bottles and tape for a small raft
- ☐ A bag of coins, all the same size
- ☐ The tub of water
- ☐ A foil tray about the same size
- ☐ A piece of cardboard the same size



Stay safe

- Keep the water tub on the floor.
- Wipe up spills so nobody slips.
- Ask an adult if you need scissors.

START HERE: BUILD



Build your floating house

1 · DO IT: TICK EACH STEP

- 1 ☐ Screw the lids on all 4 bottles, tight.
- 2 ☐ Tape the bottles side by side to make a raft.
- 3 ☐ Tape the card or tray on top. This is the floor.
- 4 ☐ Build a small LEGO house in the middle.
- 5 ☐ Put your house gently on the water.
- 6 ☐ Look from the side. Is it flat and level?

2 · LOOK & RECORD: CIRCLE WHAT YOU SEE

	Does your house float?	How many bottles did you use?	How many LEGO bricks in your house?
My floating house	☒ Yes, and level ☐ Yes, but it leans ☐ No, it sinks	bottles	bricks

 Draw your floating house

WORDS TO USE

bottles raft floor house floats level leans strong wobbly

TEST 1

Big raft or small raft?

? Will a bigger raft carry more coins?

1 · MY GUESS: CIRCLE ONE

 Small raft (2 bottles)

  Big raft (4 bottles)

 Both the same

How sure am I?



2 · DO IT: TICK EACH STEP

- ☐  Tape 2 bottles together. This is the small raft.
- ☐  Float the small raft in the tub.
- ☐  Put coins in the middle, one at a time.
- ☐  Stop when water comes on, or it tips.
- ☐  Tap the counter once for every coin.
- ☐  Now test the big raft the same way.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	 How many coins did it carry?	What happened at the end?
 Small raft	Try 1 Try 2 Try 3 Middle coins	 Water came on  It tipped over  It sank
  Big raft	Try 1 Try 2 Try 3 Middle coins	 Water came on  It tipped over  It sank

 Draw it: Small raft

 Draw it: Big raft

4 · WHAT HAPPENED? CIRCLE ONE

☒ My guess was right ☐ Partly right ☐ Something different happened ☐ Not sure yet

I found out that _____

because _____

WORDS TO USE

coins heavy sank tipped water big small more less

TEST 2 Middle or side?

? Where should heavy things go so the house does not tip?

1 · MY GUESS: CIRCLE ONE

 In the middle

 On one side

 No difference

How sure am I?   

2 · DO IT: TICK EACH STEP

- 1 ☐  Test 1: put coins in the middle, one at a time.
- 2 ☐  Stop when water comes on, or it tips.
- 3 ☐  Count the coins with the counter.
- 4 ☐  Test 2: put coins on one edge only.
- 5 ☐  Stop and count again.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	 How many coins did it carry?	What happened at the end?
 Coins in the middle	Try 1 Try 2 Try 3 Middle coins	 Water came on  It tipped over  It sank
 Coins on one edge	Try 1 Try 2 Try 3 Middle coins	 Water came on  It tipped over  It sank

 Draw it: Coins in the middle

 Draw it: Coins on one edge

4 · WHAT HAPPENED? CIRCLE ONE

 My guess was right

 Partly right

 Something different happened

 Not sure yet

I found out that _____

because _____

WORDS TO USE

middle edge side tipped balanced wobbly heavy coins

TEST 3 Best floating material

? Which raft will carry the most coins?

1 • MY GUESS: CIRCLE ONE



Plastic bottles



A foil tray



Cardboard

How sure am I?



2 • DO IT: TICK EACH STEP

- 1 ☐ Check all three rafts are about the same size.
- 2 ☐ Float one raft. Add coins in the middle.
- 3 ☐ Stop when water comes on, or it sinks.
- 4 ☐ Count the coins with the counter.
- 5 ☐ Test the other two rafts the same way.

3 • LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How many coins did it carry?	What happened at the end?	After the test, is it soggy?
Bottles	Try 1 Try 2 Try 3 Middle coins	Water came on It tipped over It sank	Still strong Soggy and soft
Foil tray	Try 1 Try 2 Try 3 Middle coins	Water came on It tipped over It sank	Still strong Soggy and soft
Cardboard	Try 1 Try 2 Try 3 Middle coins	Water came on It tipped over It sank	Still strong Soggy and soft

Draw it: Bottles

Draw it: Foil tray

Draw it: Cardboard

4 • WHAT HAPPENED? CIRCLE ONE



My guess was right



Partly right



Something different happened



Not sure yet

I found out that

because

WORDS TO USE

plastic

foil

cardboard

soggy

strong

leaked

floats

sank

Art corner: Floating village

Draw a whole floating village: houses, gardens and bridges on the water.



Maths corner: Coin count

Write your 3 tries in order: ____ ____ _____. The middle one is _____.



New words

Float: To stay on top of the water.

Raft: A flat boat made by joining things that float.

Buoyancy: The push of water that holds things up.

Fair test: Change one thing. Keep everything else the same.

Balance: When weight is spread out so nothing tips.

Material: What something is made of.

What I can do now colour the stars

I can build a raft that floats.



I can test size, balance and materials fairly.



I can use my results to design a better floating house.

